

AEGIS Security Operations Center Dashboard

Full-stack real-time SOC (Security Operations Center) dashboard for the GNS3 AEGIS-SecureCompany lab. Monitors live attacks and defenses — dashboard is monitoring-only; all real attack and defense happens on the GNS3 virtual machines.

Lab Topology (Current — v3, 2026-07-19)

[Internet / NAT cloud (virbr0)]

direct cable

[Router — MikroTik CHR]

ether1: 192.168.122.2/24 ← Internet side

ether2: 192.168.10.1/24 ← Attacker (Kali) side — DHCP server

ether3: 10.0.23.1/30 ← pfSense WAN link

direct cable

[Kali / Attacker]

eth0 → Router ether2

IP: DHCP 192.168.10.x

(no switch)

[pfSense 2.7.2]

WAN: 10.0.23.2/30

BANK_WEB: 10.10.10.1/24

CUSTOMER_DB: 10.20.20.1/24

MGMT: 10.30.30.1/24

[DMZ Zone]

[INT Zone]

[MGMT Zone]

[company-web-server]

10.10.10.10

Apache

Suricata

Fail2ban

[company-customer-db]

10.20.20.20

PostgreSQL

Suricata

Fail2ban

[company-customer-db]

10.30.30.10

Hub agent

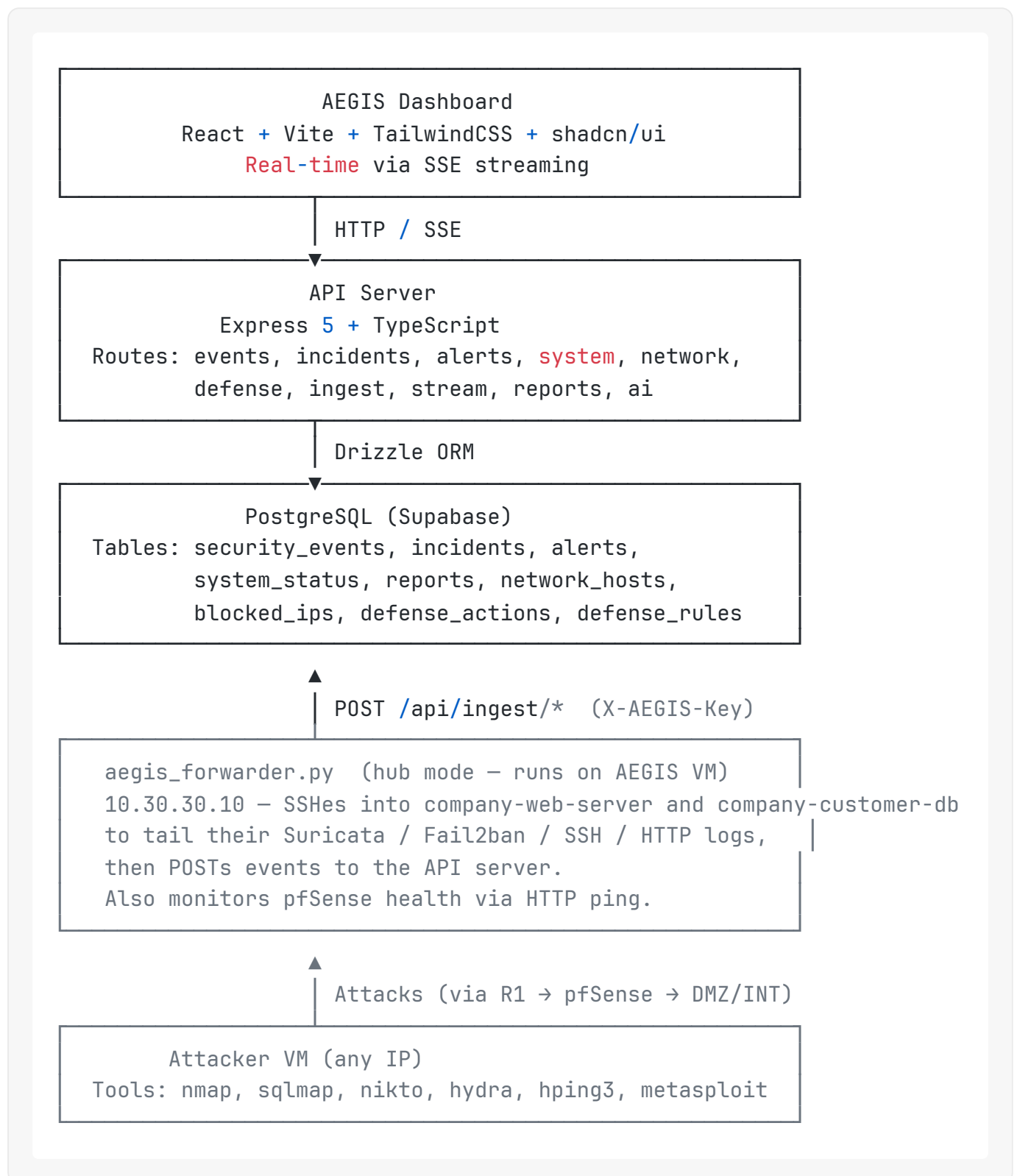
(SSH → VMs)

[aegis

⚠️ Kali IP is dynamic (DHCP 192.168.10.x) — connected directly to Router ether2, no switch. Attacker route: `sudo ip route add 10.0.0.0/8 via 192.168.10.1`

Removed from topology: R2 (MikroTik), Switch1, bank-mail server, teller-pc workstation, Cowrie honeypot.

Architecture



Features

Security Monitoring

- **Command Center** — Live dashboard with attack volume charts, event counts, critical threat indicators
- **Security Events** — Real-time event feed from all sensors (Suricata, Fail2ban, SSH monitor)
- **Incidents** — Incident management with severity tracking and status updates
- **Active Alerts** — Alert triage panel with acknowledge/resolve actions, Telegram notifications
- **System Status** — Health monitoring for all defense components (per-VM service status)

Network & Defense

- **Network Monitor** — Live topology map (pfSense, company-web-server, company-customer-db, aegis-forwarder), connected hosts table with real last-seen timestamps
- **Defense Center** — Auto defense status (Fail2ban, Suricata) per device + Admin manual IP block/unblock + pfSense WAN block via REST API

Intelligence

- **AI Analysis** — Groq llama-3.3-70b threat analysis, per-IP defense recommendations, event explanations (Burmese + English)
 - **Reports** — Auto-scheduled SOC reports (Burmese language), Telegram delivery
 - **Setup Guide** — Step-by-step hub-mode setup instructions
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Tech Stack

Layer	Technology
Frontend	React 18, Vite, TypeScript, TailwindCSS v4, shadcn/ui
Backend	Express 5, Node.js 24, TypeScript
Database	PostgreSQL + Drizzle ORM (Supabase pooler)
AI	Groq API — llama-3.3-70b-versatile
Alerts	Telegram Bot API
Validation	Zod, drizzle-zod
Real-time	Server-Sent Events (SSE)
Monorepo	pnpm workspaces
Hosting	Render (API) + Vercel (Dashboard) + Supabase (DB)

Project Structure

```
aegis-soc-dashboard/
├── artifacts/
│   ├── aegis-dashboard/           # React frontend
│   │   └── src/
│   │       ├── pages/             # dashboard, events, incidents, alerts,
│   │       │                       # system, network, defense, reports, setup
│   │       └── components/        # layout, ui components
│   └── api-server/                # Express API server
│       └── src/
│           ├── routes/            # events, incidents, alerts, system,
│           │                       # network, defense, ingest, stream, reports
│           └── lib/               # broadcaster, auto-defense, groq-client,
│                                   # telegram, scheduler, attack-tracker
├── lib/
│   ├── db/                       # Drizzle ORM schema + migrations
│   ├── api-spec/                 # OpenAPI specification
│   ├── api-client-react/         # Generated React Query hooks
│   └── api-zod/                  # Generated Zod schemas
├── scripts/
│   └── src/
│       └── aegis_forwarder.py     # Hub-mode agent (runs on AEGIS VM 10.30.10.1)
└── docs/
    ├── AEGIS_VM_SETUP.md         # AEGIS VM hub setup guide
    ├── network-architecture.md   # Full network topology & IP plan
    ├── SYSTEM_ARCHITECTURE.md    # System data flow & code structure
    ├── API.md                    # API endpoint reference
    └── PROJECT_LOG.md            # Project development log
```

Quick Start (Development)

Prerequisites

- Node.js 24+
- pnpm 9+
- PostgreSQL database (or Supabase)

Installation

```
git clone https://github.com/sohu2723-star/aegis-soc-dashboard
cd aegis-soc-dashboard
pnpm install
```

Environment Variables

```
# Required – API Server
DATABASE_URL=postgresql://user:password@localhost:5432/aegis
SESSION_SECRET=your-session-secret
AEGIS_INGEST_KEY=your-ingest-key
AEGIS_ADMIN_KEY=your-admin-key

# Optional
GROQ_API_KEY=your-groq-key           # AI analysis
TELEGRAM_BOT_TOKEN=your-bot-token    # Alert notifications
TELEGRAM_CHAT_ID=your-chat-id
```

Run

```
# API Server (port 3000)
PORT=3000 pnpm --filter @workspace/api-server run dev

# Dashboard
pnpm --filter @workspace/aegis-dashboard run dev
```

Connecting the AEGIS VM

See [docs/AEGIS_VM_SETUP.md](#) for complete step-by-step setup.

Quick summary:

1. AEGIS VM (10.30.30.10) runs `aegis_forwarder.py --mode hub`

2. Hub SSHes into company-web-server (10.10.10.10) and company-customer-db (10.20.20.20)
3. Tails their Suricata, Fail2ban, SSH, HTTP logs remotely
4. POSTs all events to Render API — dashboard updates live

Ingest API

All endpoints require `X-AEGIS-Key` header.

Endpoint	Source	Description
<code>POST /api/ingest/event</code>	Any	Generic security event
<code>POST /api/ingest/suricata</code>	Suricata	EVE JSON format
<code>POST /api/ingest/fail2ban</code>	Fail2ban	Ban/unban events
<code>POST /api/ingest/ssh</code>	auth.log	SSH login events
<code>POST /api/ingest/http</code>	ModSecurity	Web attack events
<code>POST /api/network/hosts</code>	Forwarder	Heartbeat / host registration

Defense System

Auto Defense

- SSH brute-force → iptables DROP on company-web-server
- Port scan / web attack → iptables DROP + pfSense WAN block
- Critical event → pfSense REST API block rule (persistent, applied immediately)
- All high+ alerts → Telegram notification (immediate push)

Manual Defense (Dashboard)

- Defense Center → enter any IP + reason → Block
- pfSense WAN block via REST API (if PFSENSE_API_KEY configured)
- Unblock at any time from Active Blocks list

- Full audit log of all block/unblock/pfSense actions
-

Team

Internship project — cybersecurity lab

- Red Team: Attack simulation (nmap, sqlmap, hydra, hping3, metasploit)
 - Blue Team: Defense (Suricata/Fail2ban), monitoring, incident response, AI analysis
-

License

MIT

AEGIS SOC Poster

The enhanced, presentation-ready AEGIS SOC poster is stored in the repository at <docs/assets/aegis-soc-poster-enhanced.svg>.

AEGIS

AI-POWERED CYBER ATTACK DETECTION AND AUTOMATED DEFENSE SYSTEM

DETECT • ANALYZE • RESPOND • DEFEND

THE PROBLEM / OUR SOLUTION

- Increasing attacks on critical systems
- Complex threats are difficult to detect
- Manual monitoring is slow and error-prone
- Real-time visibility is essential

AI-DRIVEN DETECTION

Log, traffic and rule-based detection

REAL-TIME MONITORING

Dashboard, threat map and SSE

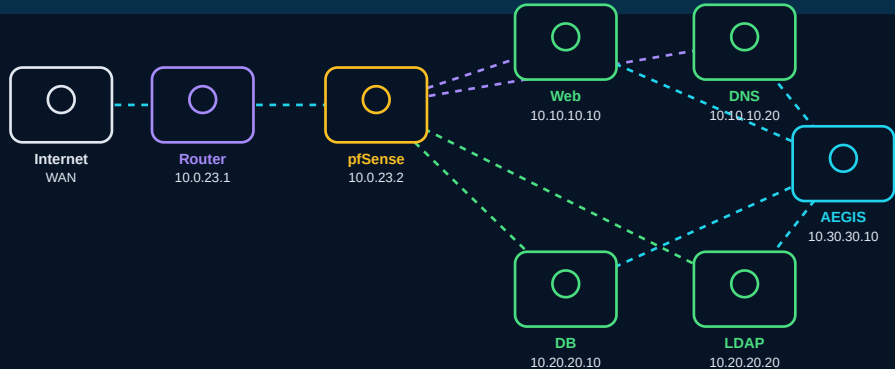
AUTOMATED DEFENSE

Safe response with rollback

ALERTS & NOTIFICATIONS

Dashboard and Telegram alerts

NETWORK TOPOLOGY



DATA COLLECTION & DETECTION PIPELINE

1 SOURCES

Suricata · fail2ban

2 FORWARDER

SSH log collection

3 API SERVER

Validate · classify

4 STORAGE

PostgreSQL · Drizzle

5 ALERT

SSE · Telegram

COMMAND CENTER DASHBOARD

TOTAL EVENTS

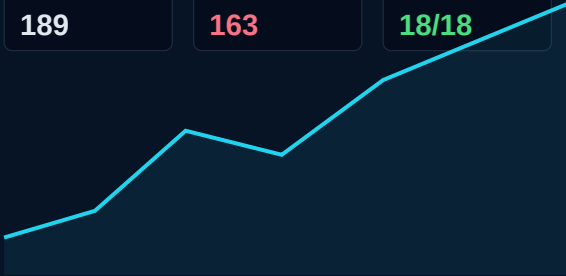
189

CRITICAL

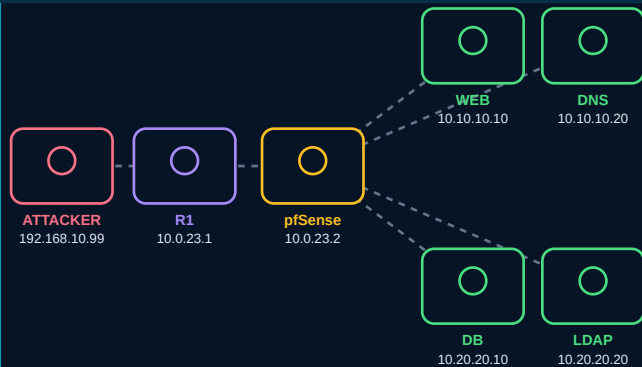
163

ONLINE

18/18



THREAT MAP — LIVE VIEW



AUTOMATED DEFENSE & RESPONSE WORKFLOW

1

1 DETECT

Security event

2

2 ANALYZE

Rule engine

3

3 REVIEW

Admin decision

4

4 SAFETY

Validate & rollback

5

5 EXECUTE

Defense agent

6

6 VERIFY

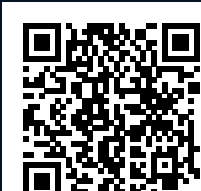
Service health

7

7 RESOLVE

Close incident

DEMO ACCESS — READ ONLY



TECHNOLOGY STACK

PERIMETER

pfSense · Suricata

DEFENSE

fail2ban · iptables

API

Node.js · Express 5

FRONTEND

React 19 · Vite · TypeScript

NETWORK

MikroTik CHR

SERVICES

Apache · BIND9 · OpenLDAP

DATABASE

PostgreSQL · Drizzle

REALTIME

SSE · Telegram

KEY BENEFITS

- ✓ Real-time threat detection
- ✓ Automated incident response
- ✓ Reduced response time
- ✓ Secure and auditable
- ✓ Scalable SQL platform

SCAN FOR DEMO

Watch 15 min. TypeScript

GoE Telegram

Get our GoE platform

AEGIS: PROTECTING YOUR INFRASTRUCTURE. DEFENDING YOUR FUTURE.